

NRES 746 – Week 5, Lecture 2: Writing the Likelihood

Name: _____

Hand this in. Like every WOD, this one is collected, so make sure your name is at the top and turn in your own copy (not a group copy) next Monday.

Instructions: This workout takes the place of today's lecture. Plan on about 30 minutes: work alone on Parts 1-3 (~15 min), check in with your group (~3 min), then work through Parts 4-5 with your group (~10 min). Please keep laptops closed and do not use AI tools for this one. Everything here is designed for pencil and paper. Calculators are fine for arithmetic, and fractions are fine as final answers. If you get stuck, note where and move on. Nobody will call time, so someone in your group should keep an eye on the clock.

Where we left off

On Monday you went *forward*: you chose a data-generating model and imagined the data it would produce. Today we go *backward*: the data are already in hand, and the question is what they tell us about the parameters. The probability formulas are the ones you already know from Monday and Lab 3. What changes is which quantity you treat as known and which you treat as unknown.

We will stay with the sage-grouse nest-fate data from Monday. Here are four nests, with outcome $y_i = 1$ if at least one egg hatched and $y_i = 0$ if the nest failed:

Nest	Sagebrush cover (%)	Outcome (1 = hatched, 0 = failed)
1	30	1
2	20	1
3	10	0
4	30	1

Part 1: One observation (individual, ~4 min)

For now, ignore cover and suppose every nest has the same unknown probability p of success.

1a. Write down the probability of each possible outcome for a single nest.

$P(y = 1 \mid p) =$ _____ $P(y = 0 \mid p) =$ _____

1b. Write one formula that gives the probability of an observed outcome y (either 0 or 1) as a function of y and p , with no “if” statements or separate cases. (The binomial pmf from Lab 3, with a single trial, is one place to start.)

Definition. The **likelihood** of a parameter value is the probability of the data you actually observed, if that parameter value were the truth. It is written as a function of the parameter, with the data held fixed: for a single observation, $\mathcal{L}(p \mid y) = P(y \mid p)$.

1c. Nest 1 hatched ($y_1 = 1$). Evaluate the likelihood of $p = 0.75$ and of $p = 0.25$ given this observation.

$\mathcal{L}(0.75 \mid y_1 = 1) = \underline{\hspace{2cm}}$ $\mathcal{L}(0.25 \mid y_1 = 1) = \underline{\hspace{2cm}}$

1d. In your formula from 1b, which symbol is the data and which is the parameter? When you did probability calculations in Lab 3, which one did you treat as known? In a likelihood, which one do you treat as known?

Part 2: Several observations (individual, ~8 min)

Now use all four nests. Assume every nest has the same success probability p , and that the nests are independent of one another.

2a. Write the likelihood of all four observations, $\mathcal{L}(p \mid y_1, y_2, y_3, y_4)$, in terms of your formula from 1b. Use product notation ($\prod$).

2b. Plug in the observed outcomes and simplify to the form $p^j(1-p)^k$. What are j and k ?

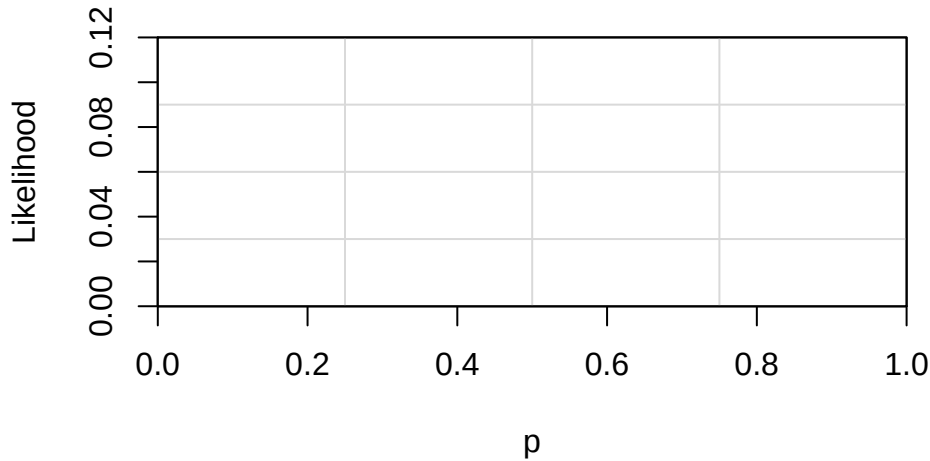
$j = \underline{\hspace{2cm}}$ $k = \underline{\hspace{2cm}}$

2c. Evaluate your answer to 2b at the three values of p below. Fractions are fine.

p	Likelihood
0.25	
0.50	
0.75	

2d. Add up your three likelihood values. Do they sum to 1? Should they? What does your answer say about how to read a likelihood value?

2e. Plot your three values on the axes below, then sketch the whole curve $\mathcal{L}(p)$ for $0 \leq p \leq 1$. Mark where you think the peak is. Is that where you would have guessed from the raw data alone?



Part 3: The log-likelihood (individual, ~3 min)

3a. Take the log of your product from 2a and use the log rules to rewrite it as a sum. (Recall that $\log(uv) = \log u + \log v$ and $\log(u^c) = c \log u$.) Write your result for a general set of n nests with outcomes $y_1, \dots, y_n$.

$$\ell(p) = \log \mathcal{L}(p \mid y_1, \dots, y_n) =$$

3b. Suppose the monitoring program had 2,000 nests and each nest contributed a likelihood of about 0.5. Roughly how big is the product of all 2,000 terms? (Use $\log_{10} 2 \approx 0.30$ and give your answer as $10^{\text{something}}$.) Standard computer arithmetic cannot store positive numbers much smaller than 10^{-308} . Why might you want to work with the sum from 3a instead? What is its value here (use $\ln 2 \approx 0.69$)?

Group check-in (~3 min)

Compare your answers to Parts 1-3 with your group. Where you differ, argue it out and revise. The sum-to-1 question (2d) and the location of the peak (2e) are good places to start. Then answer together:

Group question. Multiplying the four nest likelihoods together depended on an assumption. Name one biological reason it might fail for real sage-grouse nests.

Part 4: Adding the covariate (with your group, ~6 min)

Monday's model had success probability rising with sagebrush cover. A logistic function keeps it between 0 and 1:

$$p_i = \frac{1}{1 + e^{-(a+bx_i)}}$$

where x_i is the cover around nest i . Every nest now has its own p_i . Suppose a candidate pair of parameter values is $a = -2.2$ and $b = 0.11$. Here are the success probabilities it predicts (rounded) for our four nests:

Nest	Sagebrush cover (%)	Outcome (1 = hatched, 0 = failed)	Predicted p
1	30	1	0.75
2	20	1	0.50
3	10	0	0.25
4	30	1	0.75

4a. Check the predicted p for nest 2 (20% cover) by hand.

4b. Assuming the nests are independent, what is the likelihood of this candidate parameter pair, $\mathcal{L}(a = -2.2, b = 0.11 \mid \text{data})$? Careful with the nest that failed.

4c. Compare this to the best of your three constant- p likelihoods from 2c. Which model do the data support more, and by what factor? Does that show that cover really matters for nest success? Give at least one reason for caution.

4d. Write the likelihood for n nests in general, $\mathcal{L}(a, b \mid y_1, \dots, y_n, x_1, \dots, x_n)$, using $p_i = 1/(1 + e^{-(a+bx_i)})$. Then underline the quantities that are fixed data and circle the quantities that the likelihood is a function of.

Part 5: From math to R code, on paper (with your group, ~5 min)

Later in the course you will fit models like this by writing the likelihood as an R function and handing it to an optimizer. You will get that practice in Lab 4 and beyond. For now, we only write and read code, with no computer. The built-in `dbinom(x, size = 1, prob = p)` gives the Bernoulli probability from Part 1: x is the observed outcome (0 or 1), `size` is the number of trials per observation, and `prob` is the success probability. Setting `log = TRUE` returns the log of the probability.

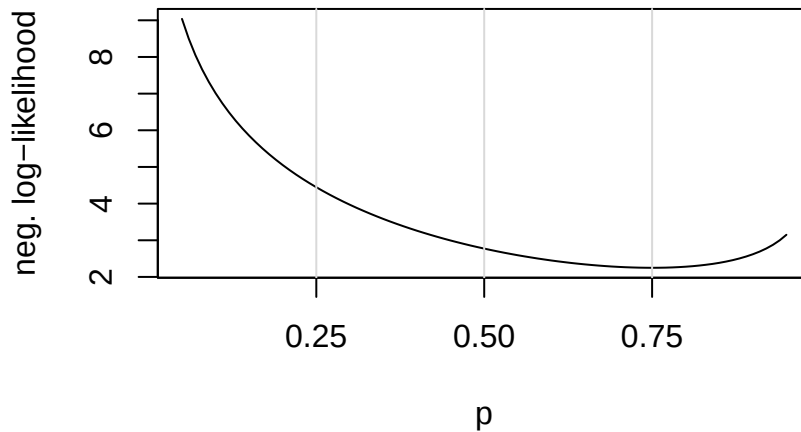
5a. Fill in the blanks so this function returns the *negative* log-likelihood of a vector of nest outcomes `y` for a single success probability `p`.

```
nll_const <- function(p, y) {  
  -sum(dbinom(x = _____, size = _____, prob = _____, log = TRUE))  
}
```

5b. With `y <- c(1, 1, 0, 1)`, what does `exp(-nll_const(0.5, y))` return? Work it out by hand. Which entry of your table in 2c should it match?

5c. A classmate ran the code below with a working `nll_const` and got this plot. (`sapply()` evaluates a function at each value of a vector in turn.)

```
p_grid <- seq(0.05, 0.95, by = 0.01)
nll_vals <- sapply(p_grid, function(p) nll_const(p, y))
plot(p_grid, nll_vals, type = "l", xlab = "p",
     ylab = "neg. log-likelihood", xaxt = "n")
axis(1, at = seq(0, 1, 0.25))
abline(v = seq(0, 1, 0.25), col = "grey85")
```



Which value of p gives the smallest negative log-likelihood? Does it agree with your sketch in 2e? Why do you think we work with the *negative* log-likelihood rather than the log-likelihood itself?

What is still fuzzy? (individual) What is the muddiest point about likelihood so far? What would you ask if this had been a lecture?

5d. Optional challenge, if you finish early. Write the covariate version, using your formula from 4d.

```
cover <- c(30, 20, 10, 30)
```

```
nll_cover <- function(par, y, cover) {  
  p <- _____  
  -sum(_____)  
}
```

If you ran `exp(-nll_cover(c(-2.2, 0.11), y, cover))`, what number should you get, approximately, and why?
